



Department of Information Technology
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch : IV Semester B.Tech IT
Course Code & Title : CS3451 & Introduction to Operating Systems
Name of the Faculty member : Mr. Balaganesh M V, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Virtual Memory

- **Course Outcome:** CO3

- **Topic Learning Outcome:** TLO9

- **Activity Chosen:** Crossword Puzzle

- **Justification:**

Virtual memory is a technique that allows the execution of processes that are not completely in memory. One major advantage of this scheme is that programs can be larger than physical memory. Further, virtual memory abstracts main memory into an extremely large, uniform array of storage, separating logical memory as viewed by the user from physical memory. In order to remember the various keywords used in virtualization, a crossword puzzle is used instead of a normal way of summary.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

- A crossword is an interesting word puzzle activity that helps the students recollect the concept and find the answer individually. It is one type of game, so this is an enjoyable activity rather than a regular multiple-choice questions test.
- The activity was conducted in a classroom for a duration of 10 minutes.
- The students were given with a crossword puzzle of 10 questions on paper. The questions were from the Input and output basics. They have to think and write the answers to the given puzzle, Figure 1.
- After the completion the answers were corrected and given to the students. Sample answers were shown in Figure:2 and 3.
- The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.

- **Crossword Puzzle Questions**

- **ACROSS**

4. Which data structure is used to implement FIFO page replacement algorithm
7. Virtual memory is an illusion of _____
8. Technique involves moving processes between main memory and a backing store is _____

9. In paging, the process is divided into _____ size pages
10. The modify bit is also called as _____
- **DOWN**
 1. In segmentation the process is divided into _____ size segments
 2. In LRU Approximation algorithm (second chance algorithm), _____ is used to find the Least recently used page
 3. _____ is the concept in which pages are copied into the main memory from the virtual memory according to the requirement
 5. When a program tries to access a page that is not loaded in the physical memory, then _____ occurs
 6. Working set model for page replacement is based on the assumption of _____

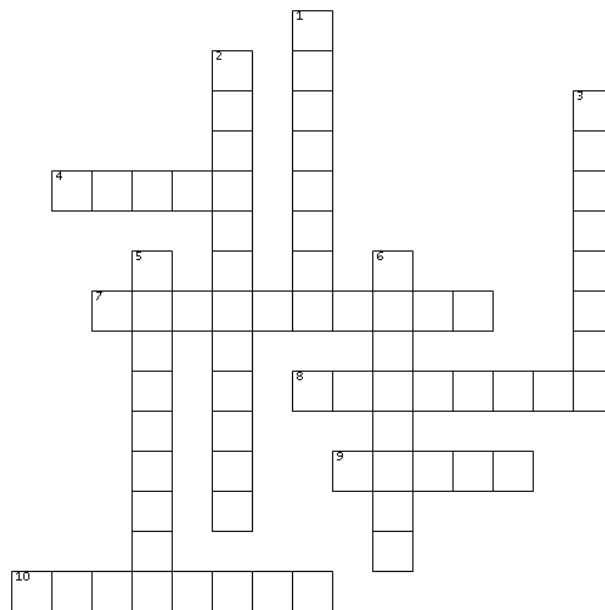


Figure1: Crossword Puzzle given to the students

- **ANSWER**

ACROSS

4. Queue
7. Mainmemory
8. swapping
9. fixed
10. dirtybit

DOWN

1. variable
2. referencebit
3. demandpaging
5. pagefault
6. locality

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO10	PSO1
CO3	3	3	1	1	1

(1 – Low

2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO10	PSO1
	3	3	2	2	1
Justification for correlation	Students will be able to understand the concept of virtual memory	Students will be able to choose the appropriate memory management technique while approaching the problem	Students will be able to provide solution for the problem by applying the concept of virtual memory.	Students communication skill will be improved as they are discussing the answers with peers	The problem solving skill earned through this activity helps the students in solving various problems

• **Images / Screenshot of the practice:**

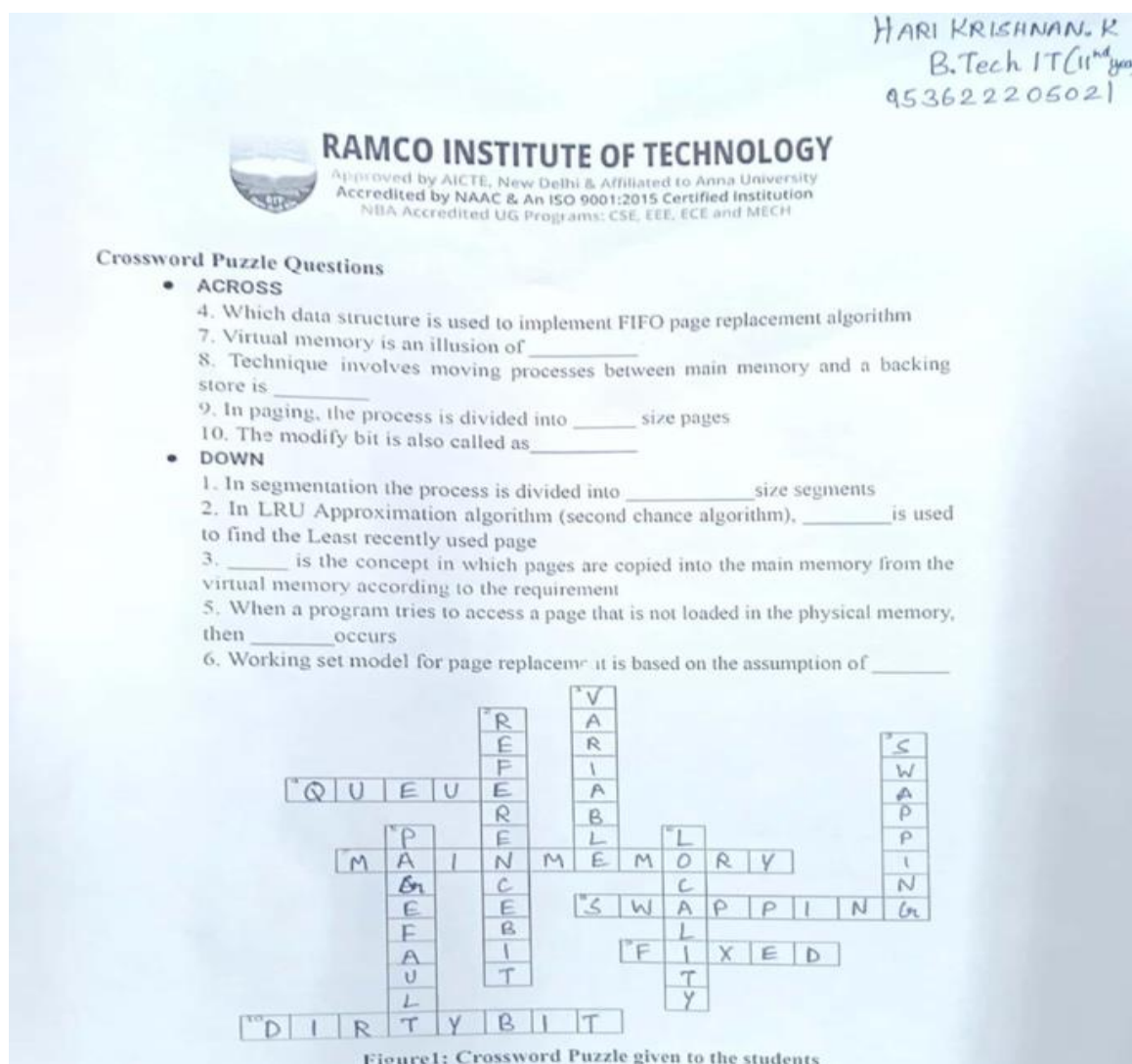


Figure 2: Puzzle solved by Harikrishnan K

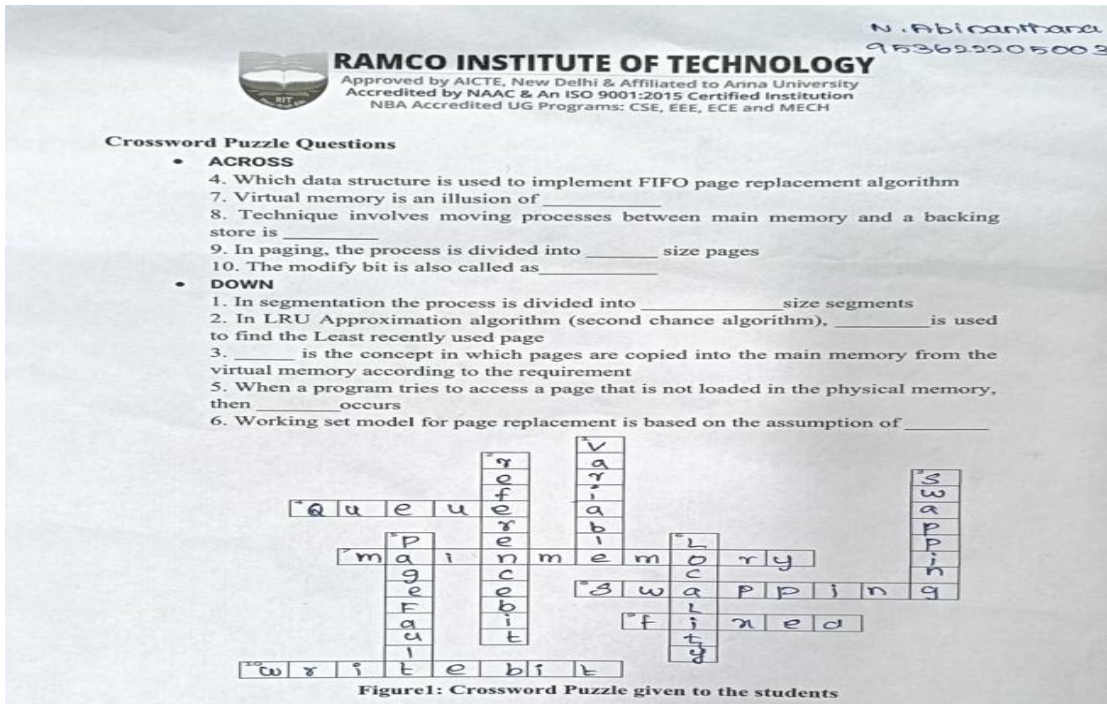


Figure 3: Crossword puzzle solved by Abinanthana N

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Some of the students find it very easy to identify the answers related to the paging technique.
- They can able to recall the concept.

❖ *Benefit of the practice:*

- The student's level of understanding of virtual memory and paging had been improved.
- They can easily identify the place where they used which type of memory allocation.

❖ *Challenges faced in implementation:*

- Some students didn't understand the working set model and struggled to find answers to the puzzle.

References:

- ❖ <https://pressbooks.gvsu.edu/nursinglessonplans/chapter/crossword-puzzle-creation/>

Signature of the faculty

HOD